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RADIATION AND MASS TRANSFER EFFECTS ON MHD FREE CONVECTION FLOW PAST A MOVING VERTICAL CYLINDER IN A POROUS MEDIUM

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Abstract:

The interaction of free convection with thermal radiation of a viscous incompressible unsteady MHD flow past a moving vertical cylinder with heat and mass transfer in a porous medium is analyzed. The fluid is a gray, absorbing-emitting but non-scattering medium and the Rosseland approximation is used to describe the radiative heat flux in the energy equation. The governing equations are solved by using an implicit finite-difference scheme of Crank-Nicolson type. The effects of various physical parameters such as thermal Grashof number, mass Grashof number, magnetic parameter, radiation parameter and Schmidt number on the velocity, temperature, concentration, local as well as average skin-friction, Nusselt number and Sherwood number for various parameters are computed and represented graphically. It is found that at small values of radiation parameter, the velocity and temperature of the fluid increases sharply near the cylinder as the time increases. Also, an increase in the magnetic field leads to a decrease in the velocity and a rise in the temperature. As the permeability parameter increases, it is seen that the flow accelerates. This model finds applications in geophysics and engineering.

Keywords: Heat and Mass transfer, MHD, Porous medium, Radiation, Finite-difference Scheme, Vertical cylinder

NOMENCLATURE

B_0	Magnetic induction	Sc	Schmidt number
c_p	specific heat at constant pressure	Sh_x	dimensionless local Sherwood number
C'	dimensional species concentration	$\overline{Sh}$	dimensionless average Sherwood number
C	dimensionless species concentration	t'	dimensional time
D	Species diffusion coefficient	t	dimensionless time
g	acceleration due to gravity	T'	temperature
Gr	thermal Grashof number	u_0	velocity of the cylinder
Gc	solutal Grashof number	u, v	velocity components in x, r directions respectively
k	thermal conductivity	U, V	dimensionless velocity components in X, R directions respectively
k_e	mean absorption coefficient	x, y	coordinates along and normal to the cylinder, respectively
K	the porous medium permeability coefficient	X, R	dimensionless coordinates along and normal to the cylinder respectively
M	magnetic parameter		
N	radiation parameter		
Nu_x	dimensionless local Nusselt number		
$\overline{Nu}$	dimensionless average Nusselt number		
Pr	Prandtl number		
R	dimensionless radial co-ordinate		
q_r	radiation heat flux		
r_0	radius of the cylinder		

Greek symbols

α	thermal diffusivity		
β	volumetric coefficient of thermal expansion	$\overline{\tau_L}$	average skin-friction
β^*	volumetric coefficient of expansion with concentration	τ_x	dimensionless local skin- friction
μ	dynamic viscosity	$\overline{\tau}$	dimensionless average skin-friction
ν	kinematic viscosity		
ρ	density of the fluid,	Subscripts	
σ	electrical conductivity	w	conditions on the wall
σ_s	Stefan-Boltzmann constant	∞	free stream conditions
τ'	Local skin-friction		

1. Introduction

The study of unsteady boundary layer flow over a moving vertical cylinder has important geophysical and engineering applications. For example, as a result of volcanic activities or tectonic movements, magmatic intrusion may occur at shallow depths in the earth crust. The intrusive magma may take the form of a cylindrical shape. An experimental and analytical study is reported by Evan et al. (1968) for transient natural convection in a vertical cylinder. Velusamy and Grag (1992), given a numerical solution for the transient natural convection over a heat generating vertical cylinder.

The interaction of heat and mass transfer with magnetic field has attracted the interest of many researches in view of its applications in MHD generators, plasma studies, nuclear reactors, geophysics and astrophysics. Michiyoshi et al. (1976) considered natural convection heat transfer from a horizontal cylinder to mercury under a magnetic field. Shanker and Kishan (1997) presented the effect of mass transfer on the MHD flow past an impulsively started infinite vertical plate. Ganesan and Rani (2000) studied the effect of MHD on unsteady free convection flow past a vertical cylinder with heat and mass transfer. Magnetic field effect on a moving vertical cylinder with constant heat flux was given by Ganesan and Loganathan (2003).

In the context of space technology and in processes involving high temperatures, the effects of radiation are of vital importance in the study of geological formations, in the exploration and thermal recovery of oil, and in the assessment of aquifers, geothermal reservoirs and underground nuclear waste storage sites. Here intrusive magma may be taken as an isothermal vertical cylinder with impulsive motion subject to radiative flux. The study of the flow along a vertical or horizontal cylinder is important in the field of geothermal power generation and drilling operations. Hossain et. al. (1999) studied Radiation-conduction interaction on mixed convection from a horizontal circular cylinder. Radiation and mass transfer effects on flow of an incompressible viscous fluid past a moving vertical cylinder was studied by Ganesan and Loganathan (2002). Mosa (1979) studied radiative heat transfer in horizontal magneto hydrodynamic channel flow with buoyancy effects and an axial temperature gradient. Radiation and mass transfer effects on two-dimensional flow past an impulsively started isothermal vertical plate were studied by Ramachandra Prasad et al. (2007).

In astrophysical regimes, the presence of planetary debris, cosmic dust etc. creates a suspended porous medium saturated with plasma fluids. Combined buoyancy-generated heat and mass transfer, due to temperature and concentration variations, in fluid-saturated porous media, has several important applications in a variety of engineering processes including heat exchanger devices, petroleum reservoirs, chemical catalytic reactors, solar energy porous water collector systems, ceramic materials(1992). Comprehensive reviews of the much of the work communicated in porous media thermal/species convection have been presented by Nield and Bejan (2006), Vafai (2005) and Ingham and Pop (2002). Yih (1999) studied the radiation effect on natural convection over a vertical cylinder embedded in porous media. More recently Nasser S. Elgazery and Hassan (2007) studied thermal radiation effect on magnetohydrodynamic (MHD) unsteady mixed-convection flow over a moving vertical cylinder with constant heat flux through a porous medium in the presence of transversal uniform magnetic field.

However, the interaction of radiation with mass transfer of an electrically conducting fluid past a vertical cylinder in a porous medium has received a little attention. Hence, the object of the present paper is to analyze

the radiation and mass transfer effects on MHD free convection flow of a viscous incompressible fluid past a vertical cylinder in the presence of porous medium. The governing boundary layer equations along with the initial and boundary conditions are first cast into a dimensionless form and the resulting system of equations then solved by an implicit finite difference scheme.

2. Mathematical Analysis

A two-dimensional unsteady hydromagnetic flow of an incompressible viscous radiating fluid past an impulsively started semi-infinite vertical cylinder of radius r_0 in a porous medium is considered.

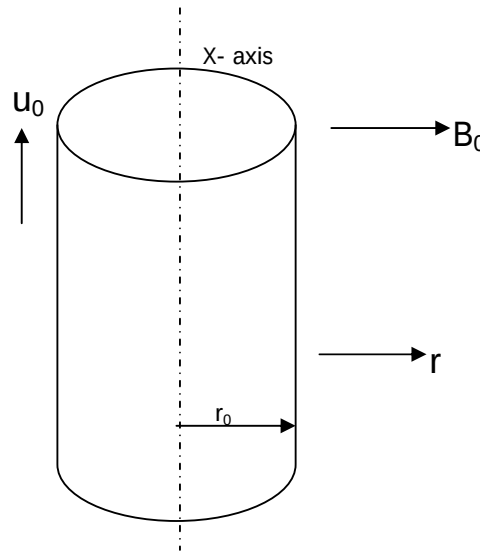


Figure.1 Physical model of the problem

Here the x -axis is taken along the axis of cylinder in the vertical direction and the radial coordinate r is taken normal to the cylinder. This is shown in the figure.1. Initially, the fluid and the cylinder are of same temperature T'_∞ and the concentration C'_∞ . At a time $t' \geq 0$, the cylinder starts moving in the vertical direction with velocity u_0 . At a later time, the surface of cylinder is raised to a uniform temperature T'_w and concentration C'_w and are maintained constantly thereafter. A uniform magnetic field is applied in the transverse direction to the cylinder. The fluid is assumed to be slightly conducting, and hence the magnetic Reynolds number is much less than unity and the induced magnetic field is negligible in comparison with the transverse applied magnetic field. The viscous dissipation is also assumed to be neglected in the energy equation due to slow motion of the cylinder. It is assumed that the concentration C' of the diffusing species in the binary mixture is very less in the comparison to the other chemical species, which are present, and hence the Soret and Dufour effects are negligible. All the physical properties are assumed to be constant except for the density in the buoyancy term. Then, under these assumptions and the Boussinesq's approximation, the flow is governed by the following system of equations:

Continuity equation

$$\frac{\partial(ru)}{\partial x} + \frac{\partial(rv)}{\partial r} = 0 \quad (1)$$

Momentum equation

$$\frac{\partial u}{\partial t'} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial r} = g\beta (T' - T'_\infty) + g\beta' (C' - C'_\infty) + \frac{\nu}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u}{\partial r} \right) - \frac{\sigma B_0^2}{\rho} u - \frac{\nu}{K'} u \quad (2)$$

Energy equation

$$\frac{\partial T'}{\partial t'} + u \frac{\partial T'}{\partial x} + v \frac{\partial T'}{\partial r} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T'}{\partial r} \right) - \frac{1}{\rho c_p} \frac{1}{r} \frac{\partial}{\partial r} (r q_r) \quad (3)$$

Mass diffusion equation

$$\frac{\partial C'}{\partial t'} + u \frac{\partial C'}{\partial x} + v \frac{\partial C'}{\partial r} = \frac{D}{r} \frac{\partial}{\partial r} \left(r \frac{\partial C'}{\partial r} \right) \quad (4)$$

The initial and boundary conditions are:

$$\begin{aligned} t' \leq 0: \quad u=0, \quad v=0, \quad T'=T'_\infty, \quad C'=C'_\infty \quad \text{for all } x \geq 0 \text{ and } r \geq 0 \\ t' > 0: \quad u=u_0, \quad v=0, \quad T'=T'_w, \quad C'=C'_w \quad \text{at } r=r_0 \\ u=0, \quad T'=T'_\infty, \quad C'=C'_\infty \quad \text{at } x=0 \\ u \rightarrow 0, T' \rightarrow T'_\infty, \quad C' \rightarrow C'_\infty \quad \text{as } r \rightarrow \infty \end{aligned} \quad (5)$$

We now assume Rosseland approximation (Brewster [13]), which leads to the radiative heat flux q_r is given by

$$q_r = -\frac{4\sigma_s}{3k_e} \frac{\partial T'^4}{\partial y} \quad (6)$$

It should be noted that by using the Rosseland approximation, the present analysis is limited to optically thick fluids. If temperature differences within the flow are sufficiently small, then Equation (6) can be linearized by expanding T'^4 into the Taylor series about T'_∞ , which after neglecting higher order terms takes the form

$$T'^4 \cong 4T_\infty'^3 T' - 3T_\infty'^4 \quad (7)$$

In view of Equations (6) and (7), Equation (3) reduces to

$$\frac{\partial T'}{\partial t'} + u \frac{\partial T'}{\partial x} + v \frac{\partial T'}{\partial r} = \frac{\alpha}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T'}{\partial r} \right) + \frac{16\sigma_s T_\infty'^3}{3k_e \rho c_p} \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T'}{\partial r} \right) \quad (8)$$

In order to write the governing equations and the boundary conditions in dimensionless form, the following non-dimensional quantities are introduced.

$$\begin{aligned} X = \frac{x\nu}{u_0 r_0^2}, \quad R = \frac{r}{r_0}, \quad t = \frac{t'\nu}{r_0^2}, \quad U = \frac{u}{u_0}, \quad V = \frac{v r_0}{\nu}, \quad Gr = \frac{g\beta r_0^2 (T'_w - T'_\infty)}{\nu u_0}, \\ Gc = \frac{g\beta^* r_0^2 (C'_w - C'_\infty)}{\nu u_0}, \quad N = \frac{4\sigma_s T_\infty'^3}{k_e k}, \quad M = \frac{\sigma B_0^2 r_0^2}{\rho \nu}, \quad T = \frac{T' - T'_\infty}{T'_w - T'_\infty} \\ C = \frac{C' - C'_\infty}{C'_w - C'_\infty}, \quad Pr = \frac{\nu}{\alpha}, \quad Sc = \frac{\nu}{D}, \quad K = \frac{K'}{r_0^2} \end{aligned} \quad (9)$$

Equations (1), (2), (8) and (4) are reduced to the following non-dimensional form

$$\frac{\partial(RU)}{\partial X} + \frac{\partial(RV)}{\partial R} = 0 \quad (10)$$

$$\frac{\partial U}{\partial t} + U \frac{\partial U}{\partial X} + V \frac{\partial U}{\partial R} = GrT + GcC + \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial U}{\partial R} \right) - \left(M + \frac{1}{K} \right) U \quad (11)$$

$$\frac{\partial T}{\partial t} + U \frac{\partial T}{\partial X} + V \frac{\partial T}{\partial R} = \frac{1}{Pr} \left(1 + \frac{4}{3} N \right) \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial T}{\partial R} \right) \quad (12)$$

$$\frac{\partial C}{\partial t} + U \frac{\partial C}{\partial X} + V \frac{\partial C}{\partial R} = \frac{1}{Sc} \frac{1}{R} \frac{\partial}{\partial R} \left(R \frac{\partial C}{\partial R} \right) \quad (13)$$

The corresponding initial and boundary conditions are

$$t \leq 0: U=0, V=0, T=0, C=0$$

$$t > 0: U=1, V=0, T=1, C=1 \quad \text{at } R=1$$

$$U=0, T=0, C=0 \quad \text{at } X=0 \quad (14)$$

$$U \rightarrow 0, T \rightarrow 0, C \rightarrow 0 \quad \text{as } R \rightarrow \infty$$

Knowing the velocity, temperature and concentration fields, it is interesting to study the local and average skin-friction, Nusselt number and Sherwood numbers are defined as follows.

Local and average skin-friction in non-dimensional form are given by

$$\tau_x = \frac{\tau'}{\rho u_0^2} = - \left(\frac{\partial U}{\partial R} \right)_{R=1} \quad (15)$$

$$\bar{\tau} = - \int_0^1 \left(\frac{\partial U}{\partial R} \right)_{R=1} dX \quad (16)$$

Local and average Nusselt number in non-dimensional form are given by

$$Nu_x = -X \left(\frac{\partial T}{\partial R} \right)_{R=1} \quad (17)$$

$$\bar{Nu} = - \int_0^1 \left(\frac{\partial T}{\partial R} \right)_{R=1} dX \quad (18)$$

Local and average Sherwood number in non-dimensional form are given by

$$Sh_x = -X \left(\frac{\partial C}{\partial R} \right)_{R=1} \quad (19)$$

$$\bar{Sh} = - \int_0^1 \left(\frac{\partial C}{\partial R} \right)_{R=1} dX \quad (20)$$

3. Numerical Technique

In order to solve the unsteady, non-linear coupled Equations (10) to (13) under the conditions (14), an implicit finite difference scheme of Crank-Nicolson type has been employed. The region of integration is considered as a rectangle with sides $X_{\max}(=1)$ and $R_{\max}(=14)$, where $R_{\max}$ corresponds to $R = \infty$, which lies very well outside the momentum, energy and concentration boundary layers. The maximum of R was chosen as 14 after some preliminary investigations, so that the last two of the boundary conditions (14) are satisfied within the tolerance limit 10^{-5} .

The finite difference equations corresponding to Equations (10) to (13) are as follows:

$$\frac{[U_{i,j}^{n+1} - U_{i-1,j}^{n+1} + U_{i,j}^n - U_{i-1,j}^n + U_{i,j-1}^{n+1} - U_{i-1,j-1}^{n+1} + U_{i,j-1}^n - U_{i-1,j-1}^n]}{4\Delta X} + \frac{[V_{i,j}^{n+1} - V_{i,j-1}^{n+1} + V_{i,j}^n - V_{i,j-1}^n]}{2\Delta R} + \frac{V_{i,j}^{n+1}}{1 + (j-1)\Delta R} = 0 \quad (21)$$

$$\frac{[U_{i,j}^{n+1} - U_{i,j}^n]}{\Delta t} + U_{i,j}^n \frac{[U_{i,j}^{n+1} - U_{i-1,j}^{n+1} + U_{i,j}^n - U_{i-1,j}^n]}{2\Delta X} + V_{i,j}^n \frac{[U_{i,j+1}^{n+1} - U_{i,j-1}^{n+1} + U_{i,j+1}^n - U_{i,j-1}^n]}{4\Delta R}$$

$$= Gr \frac{[T_{i,j}^{n+1} + T_{i,j}^n]}{2} + Gc \frac{[C_{i,j}^{n+1} + C_{i,j}^n]}{2} + \frac{[U_{i,j-1}^{n+1} - 2U_{i,j}^{n+1} + U_{i,j+1}^{n+1} + U_{i,j-1}^n - 2U_{i,j}^n + U_{i,j+1}^n]}{2(\Delta R)^2} + \frac{[U_{i,j+1}^{n+1} - U_{i,j-1}^{n+1} + U_{i,j+1}^n - U_{i,j-1}^n]}{4[1 + (j-1)\Delta R]\Delta R} - \left(M + \frac{1}{K}\right) \frac{[U_{i,j}^{n+1} + U_{i,j}^n]}{2} \quad (22)$$

$$\frac{[T_{i,j}^{n+1} - T_{i,j}^n]}{\Delta t} + U_{i,j}^n \frac{[T_{i,j}^{n+1} - T_{i-1,j}^{n+1} + T_{i,j}^n - T_{i-1,j}^n]}{2\Delta X} + V_{i,j}^n \frac{[T_{i,j+1}^{n+1} - T_{i,j-1}^{n+1} + T_{i,j+1}^n - T_{i,j-1}^n]}{4\Delta R} = \frac{1}{Pr} \left(1 + \frac{4}{3}N\right) \frac{[T_{i,j-1}^{n+1} - 2T_{i,j}^{n+1} + T_{i,j+1}^{n+1} + T_{i,j-1}^n - 2T_{i,j}^n + T_{i,j+1}^n]}{2(\Delta R)^2} + \frac{[T_{i,j+1}^{n+1} - T_{i,j-1}^{n+1} + T_{i,j+1}^n - T_{i,j-1}^n]}{4Pr[1 + (j-1)\Delta R]\Delta R} \quad (23)$$

$$\frac{[C_{i,j}^{n+1} - C_{i,j}^n]}{\Delta t} + U_{i,j}^n \frac{[C_{i,j}^{n+1} - C_{i-1,j}^{n+1} + C_{i,j}^n - C_{i-1,j}^n]}{2\Delta X} + V_{i,j}^n \frac{[C_{i,j+1}^{n+1} - C_{i,j-1}^{n+1} + C_{i,j+1}^n - C_{i,j-1}^n]}{4\Delta R} = \frac{1}{Sc} \frac{[C_{i,j-1}^{n+1} - 2C_{i,j}^{n+1} + C_{i,j+1}^{n+1} + C_{i,j-1}^n - 2C_{i,j}^n + C_{i,j+1}^n]}{2(\Delta R)^2} + \frac{[C_{i,j+1}^{n+1} - C_{i,j-1}^{n+1} + C_{i,j+1}^n - C_{i,j-1}^n]}{4Sc[1 + (j-1)\Delta R]\Delta R} \quad (24)$$

Here, the subscript i - designates the grid point along the X - direction, j - along the R - direction and the superscript n along the t - direction. An appropriate mesh size considered for the calculation is $\Delta X = 0.05$, $\Delta R = 0.25$, and time step $\Delta t = 0.01$. During any one-time step, the coefficients $U_{i,j}^n$ and $V_{i,j}^n$ appearing in the difference equations are treated as constants. The values of U , V , T and C are known at all grid points at $t = 0$ from the initial conditions. The computations of U , V , T and C at time level $(n+1)$ using the known values at previous time level (n) are calculated as follows.

The finite difference Equation (24) at every internal nodal point on a particular i - level constitute a tri-diagonal system of equations. Such a system of equations is solved by Thomas algorithm as described in Carnahan et al. [1969]. Thus, the values of C are found at every nodal point on a particular i at $(n+1)^{th}$ time level. Similarly, the values of T are calculated from the Equation (23). Using the values of C and T at $(n+1)^{th}$ time level in the Equation (22), the values of U at $(n+1)^{th}$ time level are found in a similar manner. Thus the values of C , T and U are known on a particular i - level. The values of V are calculated explicitly using the Equation (21) at every nodal point on a particular i - level at $(n+1)^{th}$ time level. This process is repeated for various i - levels. Thus, the values of C, T, U and V are known at all grid points in the rectangular region at $(n+1)^{th}$ time level.

Computations are carried out till the steady state is reached. The steady state solution is assumed to have been reached, when the absolute difference between the values of U as well as temperature T and concentration C at two consecutive time steps are less than 10^{-5} at all grid points. The derivatives involved in the Equations (15) to (20) are evaluated using five-point approximation formula and the integrals are evaluated using Newton-Cotes closed integration formula. The truncation error in the finite-difference approximation is $O(\Delta t^2 + \Delta R^2 + \Delta X)$ and it tends to zero as $\Delta X, \Delta R, \Delta t \rightarrow 0$. Hence the scheme is compatible. The finite-difference scheme is unconditionally stable as explained by Ganesan and Rani (2000). Stability and compatibility ensures convergence.

4. Results and Discussion

In order to get a physical insight into the problem, a representative set of numerical results is shown graphically in Figs. 2-10, to illustrate the influence of physical quantities such as radiation parameter N , thermal Grashof number Gr , mass Grashof number Gc , magnetic field parameter M , permeability coefficient K and Schmidt number Sc on velocity, temperature and concentration. All computations are carried out for $Pr = 0.71$ (i.e., for air) and the corresponding values of Sc are chosen such that they represent water vapor (0.6) and carbon dioxide (0.94).

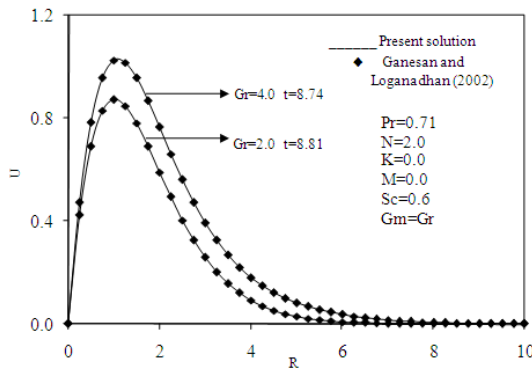


Fig. 2: Comparison of velocity profiles at X=1.0

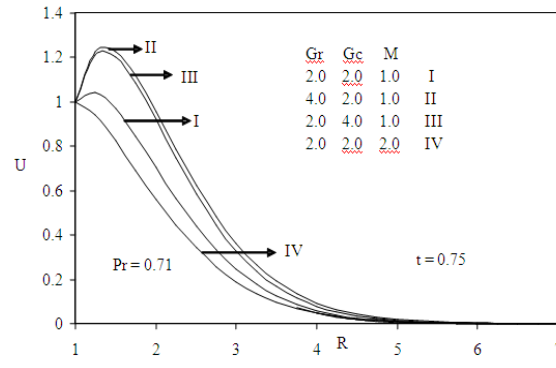


Fig.2(a): Transient velocity profiles at X=1.0 for different Gr, Gc and M

In order to ascertain the accuracy of the numerical results, the present study is compared with the previous study. The velocity profiles for $Gr=2.0, 4.0, Pr = 0.71, N = 2.0, K=0.0, M = 0.0, Sc=0.6, Gr=Gm, X = 1.0$ are compared with the available solution of Ganesan and Loganathan (2002) Fig.2. It is observed that the present results are in good agreement with that of Ganesan and Loganathan (2002).

The transient velocity profiles for different values of Gr, Gc , and M at a particular time $t = 0.75$ are shown in Fig.2(a). The thermal Grashof number signifies the relative effect of the thermal buoyancy (due to density differences) force to the viscous hydrodynamic force in the boundary layer flow. The positive values of Gr correspond to cooling of the cylinder surface by natural convection. Heat is therefore conducted away from the vertical cylinder into the fluid which increases temperature and thereby enhances the buoyancy force. It is observed that the transient velocity accelerates due to enhancement in the thermal buoyancy force, i.e., free convection effects. The solutal Grashof number Gc defines the ratio of the species buoyancy force to the viscous hydrodynamic force. It is noticed that the transient velocity increases considerably with a rise in the species buoyancy force. In both the cases it is interesting to note that as Gr or Gc increases, there is rapid rise in the velocity near the surface of vertical cylinder and then descends smoothly to the free stream velocity. As expected, the transient velocity decreases with an increase in the magnetic parameter M . It is because that the application of transverse magnetic field will result a resistive type force (Lorentz force) similar to drag force which tends to resist the fluid flow and thus reducing its velocity. Also, the boundary layer thickness decreases with an increase in the magnetic parameter.

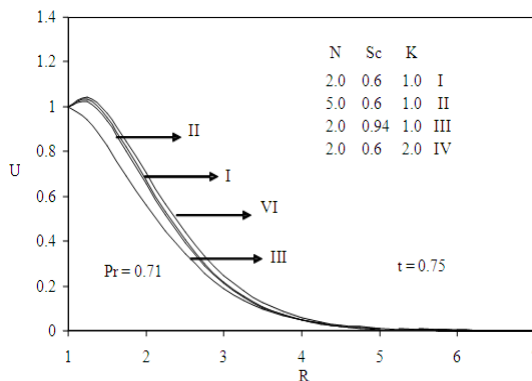


Fig.2(b): Transient velocity profiles at X=1.0 for different N, Sc and K

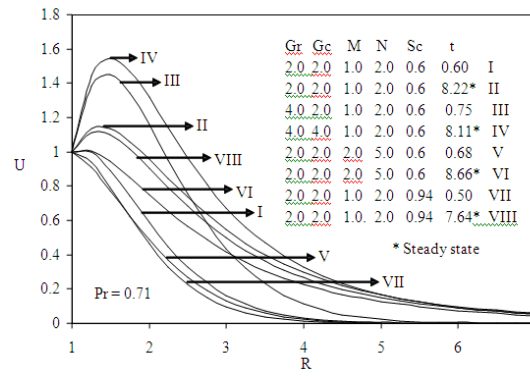


Fig.2(c): Velocity profiles at X=1.0 for different Gr, Gc, M, N and Sc

In Fig.2 (b) the transient velocity profiles for different values of N, Sc and K at a particular time $t = 0.75$ are shown. The radiation parameter N (i.e., Stark number) defines the relative contribution of conduction heat

transfer to thermal radiation transfer. An increase in N corresponds to an increase in the relative contribution of conduction heat transfer to thermal radiation heat transfer. It can be seen that as N increases from $N=2$ to $N=5$ considerable reduction is observed in the velocity. The Schmidt number Sc signifies the relative effectiveness of momentum and mass transport by diffusion in the velocity and concentration (species) boundary layers. It is observed that as the Schmidt number increases the velocity decreases. The permeability co-efficient stimulates the effect of bulk matrix impedance due to porous medium fibers. A rise in the permeability coefficient enhances the velocity of the fluid.

In Fig.2(c), the transient and steady state velocity profiles are presented for different values Gr , Gc , M , N and Sc . The steady state velocity increases with an increase in Gr or Gc . It can be seen that an increase in the thermal or species buoyancy force, reduces the time to reach the steady state. The steady state velocity decreases with an increase in M or N or Sc . The time taken to reach the steady state velocity increases as M or N increases.

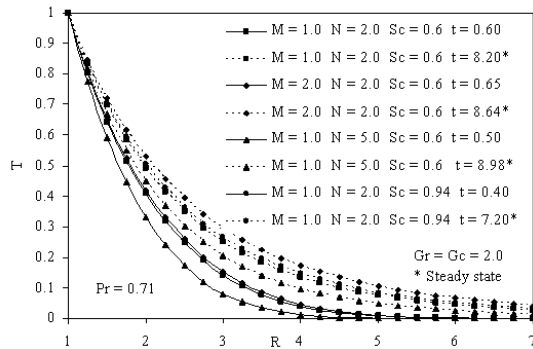


Fig.3: Temperature profiles at $X=1.0$ for different M , N and Sc

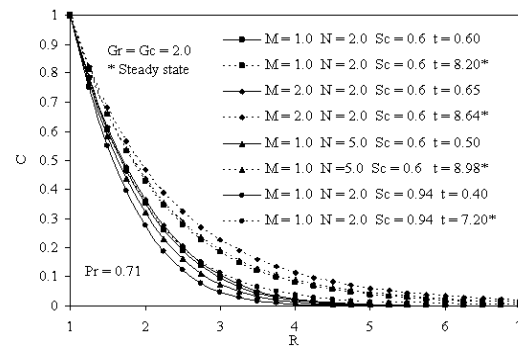


Fig. 4: Concentration profiles at $X=1.0$ for different M , N and Sc

The transient and steady state temperature profiles are presented for different values M , N and Sc in Fig.3. It is observed that, as N decreases from 5.0 to 2.0, the temperature increases markedly throughout the length of the cylinder. As a result the thermal boundary layer thickness is decreasing due to a rise in N values. It is noticed that the temperature decreases with an increase in M while it increases with an increase in Sc . The time required to reach the steady state temperature increases with an increase in M or N and it decreases with an increase in Sc .

In Fig.4, the transient and steady state concentration profiles are presented for different values of M , N and Sc . It is found that the concentration decreases as the radiation parameter N or the Schmidt number Sc increases, while it increases with an increase in M . The time required to reach the steady state concentration increases with an increase in M or N and it decreases with an increase in Sc .

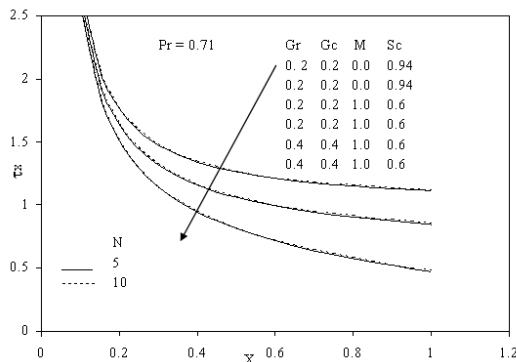


Fig. 5: Local skin friction

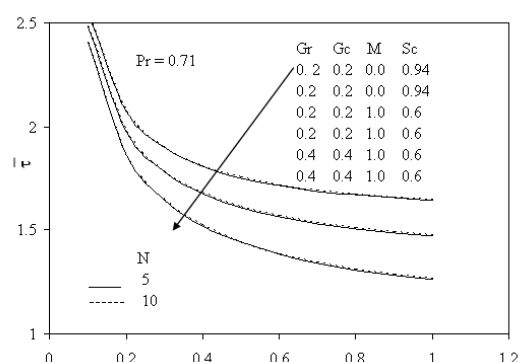


Fig. 6: Average skin friction

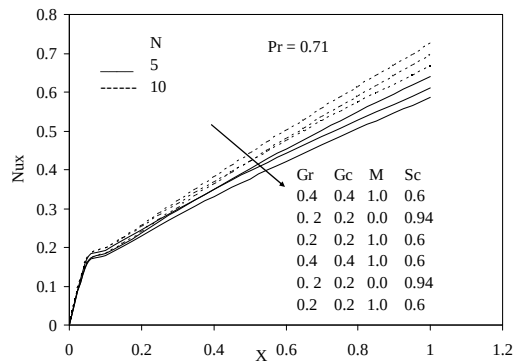


Fig. 7: Local Nusselt number

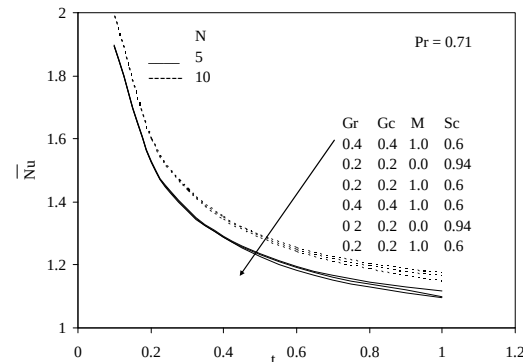


Fig. 8: Average Nusselt number

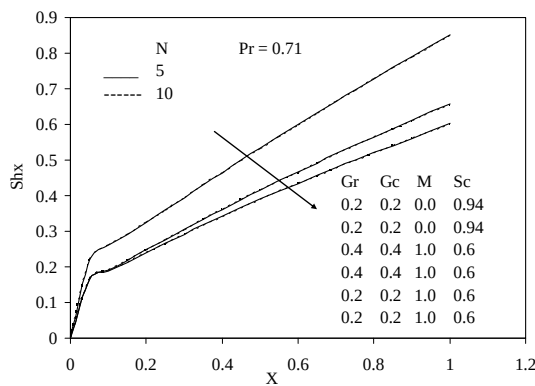


Fig. 9: Local Sherwood number

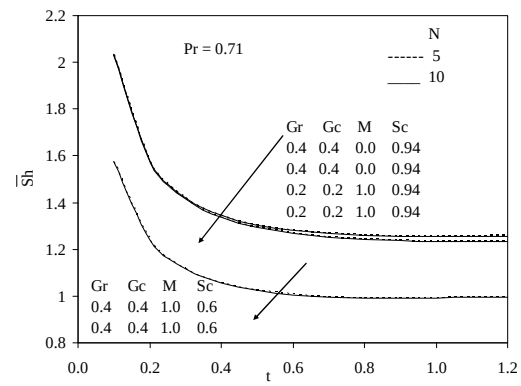


Fig. 10: Average Sherwood number

Steady-state local skin-friction (τ_x) values against the axial coordinate X are plotted in Fig.5. The local shear stress τ_x increases with an increase in Sc , while it decreases with an increase in Gr or Gc or M . The average skin-friction ($\bar{\tau}$) values are shown in Fig. 6. It is found that the average skin-friction increases with an increase in Sc , while it decreases with an increase in Gr or Gc or M , throughout the transient period. It is also observed that the average skin-friction increases as the radiation interaction parameter N increases. The local Nusselt number (Nu_x) is shown in Fig.7. The local heat transfer rate decreases with an increase in Sc , while it increases with an increase in Gr or Gc or M . Also it is found that as the radiation parameter N increases, the local Nusselt number increases. The average Nusselt number ($\bar{Nu}$) values are shown in Fig.8. It is observed that the average Nusselt number increases with an increase in Gr or Gc or N . The local Sherwood number Sh_x is plotted in Fig.9. It is noticed that Sh_x increases with an increase in Sc , where as it decreases with an increase in Gr or Gc or N . The average Sherwood number ($\bar{Sh}$) values are shown in Fig.10. It can be seen that the average Sherwood number increases with an increase in Gr or Gc or Sc , while it decreases with an increase in M .

5. Conclusion

A numerical study has been carried out to study the radiation and mass transfer effects on MHD flow of an incompressible viscous fluid past a moving vertical cylinder in a porous medium. The fluid is gray, absorbing-emitting but non-scattering medium and the Rosseland approximation is used to describe the radiative heat flux in the energy equation. A family of governing partial differential equations are solved by an implicit finite difference scheme of Crank-Nicolson type, which is stable and convergent. The results are obtained for different values of radiation parameter N , thermal Grashof number Gr , solutal Grashof number Gc , magnetic field parameter M , and Schmidt number Sc . Conclusions of this study are as follows:

1. The time required for velocity to reach the steady-state increases as radiation parameter N increases.
2. As small values of the radiation parameter N , the velocity and temperature of fluid increases sharply near the cylinder as the time t increases, which is totally absent in the absence of radiation effects.
3. The transient velocity increases with the increase in the permeability co- efficient K .
4. The skin-friction decreases with an increase M and increases with the increasing value of radiation parameter N and Schmidt number Sc .
5. The average Nusselt number increases with the increasing value of the radiation parameter.
6. The average Sherwood number increases as Gr or Gc and Sc increases.

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